

Blood Transfusion Compatibility Guide

A Comprehensive Reference for ABO and Rh Blood Groups

When it comes to blood transfusions, compatibility depends entirely on markers called **antigens** on the surface of your red blood cells. Your immune system is hardwired to attack any incoming blood that has antigens it doesn't recognize, which can trigger a severe, life-threatening immune response.

The Two Key Systems: Clinical compatibility is dictated primarily by the **ABO system** (determining Types A, B, AB, and O) and the **Rh factor** (determining whether you are Positive + or Negative -).

Quick Reference Compatibility Matrix

The table below provides a direct mapping of safe donor and recipient combinations based on standard red blood cell compatibility patterns.

Blood Type	Can Receive From	Can Donate To
O-	O-	All blood types Universal Donor
O+	O-, O+	O+, A+, B+, AB+
A-	O-, A-	A-, A+, AB-, AB+
A+	O-, O+, A-, A+	A+, AB+
B-	O-, B-	B-, B+, AB-, AB+
B+	O-, O+, B-, B+	B+, AB+
AB-	O-, A-, B-, AB-	AB-, AB+
AB+	All blood types Universal Recipient	AB+

The Golden Rules of Compatibility

To fundamentally understand why these rules exist, keep two core biological principles in mind:

1. The Rh Factor Rule

Rh-positive blood is adaptive; individuals can safely receive blood from both positive and negative donors. Conversely, Rh-negative individuals have an immune system that strictly rejects Rh-positive blood, meaning they can **only** receive Rh-negative blood units.

2. The Universal Players

O Negative (O–): This blood type completely lacks A, B, and Rh antigens on its red cell surfaces. Because there are no foreign markers for a recipient's immune system to recognize, it can be safely given to any patient. This makes it the critical *Universal Donor* used in trauma scenarios when a patient's blood type cannot be immediately tested.

AB Positive (AB+): These cells naturally express every single primary antigen (A, B, and the Rh factor). Because the patient's immune system recognizes all of these markers as "self," an AB+ individual is the *Universal Recipient* and can safely accept red blood cells from any blood group without complication.